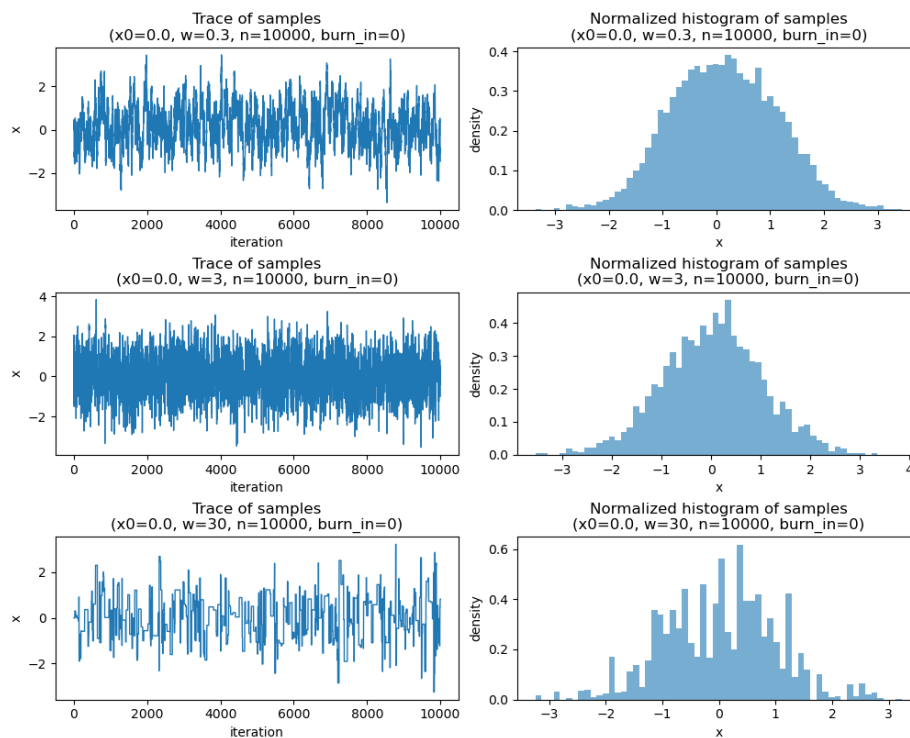


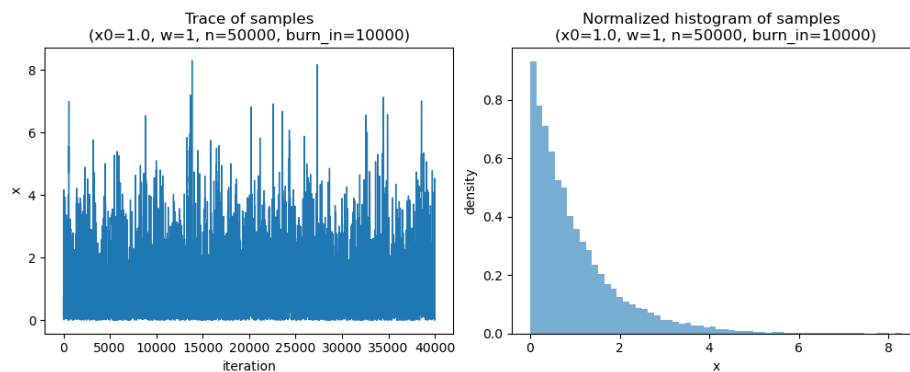
Compute 1

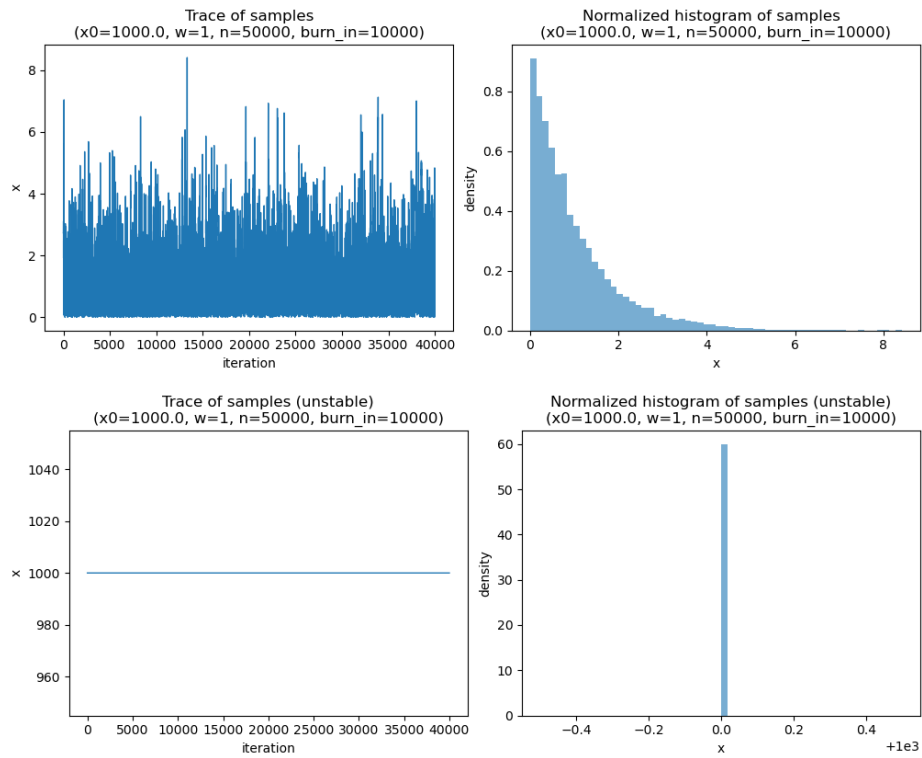
Exercise 1 and 2



The most efficient mixing seems to be with proposal window $w = 3$.

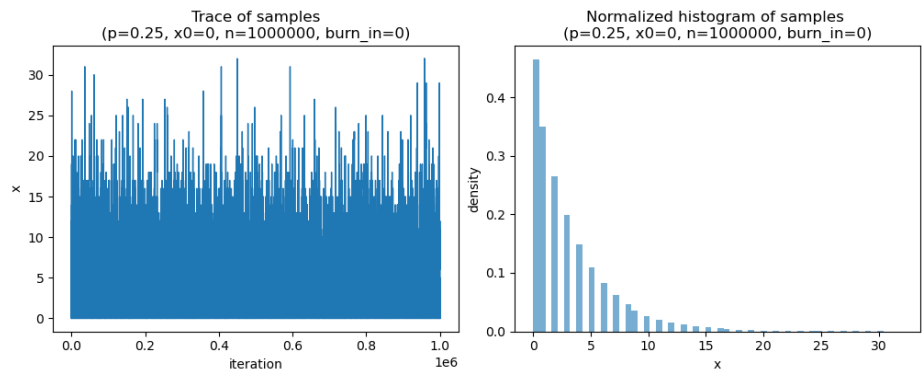
Exercise 3





In the numerically unstable version `mcexp_unstable`, the proposals are always rejected. This is because for $x_0=1000$ the numerator and denominator in the ratio both underflow to $0.0/0.0$ which evaluates to `NaN`, then `np.minimum(1.0, NaN)` returns `NaN`, and any comparison with `NaN` returns `false`.

Exercise 4



We obtain a sample mean of 3.0304 which is close to the true mean of 3.